oblig 6

oppg 1

massestrøm for R-134a $(\dot{m}_R) = 5\,[kg/min] = \frac{5}{60}\,[kg/s]$

startstrykk R-134a $(P_{1R}) = 900 \cdot 10^3\,[Pa]$

starttemp R-134a $(T_{1R}) = (60 + 273{,}15)\,[k]$

utgangstilstand R-134a: mettet væske ved $900 \cdot 10^3\,[Pa]$ $(h_2) = h_f$, ved $900 \cdot 10^5\,[Pa]$

kjølevann inn $(P_{w_1}) = 300 \cdot 10^3\,[Pa]$, $(T_{w_1}) = (10 + 275{,}15)\,[k]$

ut $(P_{w_2}) = 300 \cdot 10^3\,[Pa]$, $(T_{w_2}) = (20 + 275{,}15)\,[k]$

spesifikk varmekapasitet for vann $(C_{p_w}) = 4{,}18$ $\left[kJ/(kg\,k)\right]$

$$\dot{Q} = \dot{m}_R (h_1 - h_2) = \dot{m}_w \cdot C_{p_w} (T_{w_2} - T_{w_1})$$

$h_1 = 101{,}62\,[kJ/kg]$

$h_2 = 255{,}15\,[kJ/kg]$

$$\dot{m}_w = \frac{\dot{m}_R (h_1 - h_2)}{C_{p_w} (T_{w_2} - T_{w_1})} = \frac{\frac{5}{60}(255{,}15 - 101{,}62)}{4{,}18\,(20 - 10)} = \frac{645}{1672} \approx 0{,}38\,[kg/s]$$

oppg 2

tilstand for ventil:

$(p_1) = 1 \cdot 10^6 \, [Pa]$

$(T_1) = 36 [C]$

$(\dot{V}_1) = 1,5 \left(10^{-1}\right)^3 \, [m^3/s]$

tilstand etter ventil:

$(p_2) = 120 \cdot 10^3 \, [Pa]$

$\dot{m} = V_1 \dot{V}_1$

$p_{sat@36C} = 912,35 \, [kPa]$

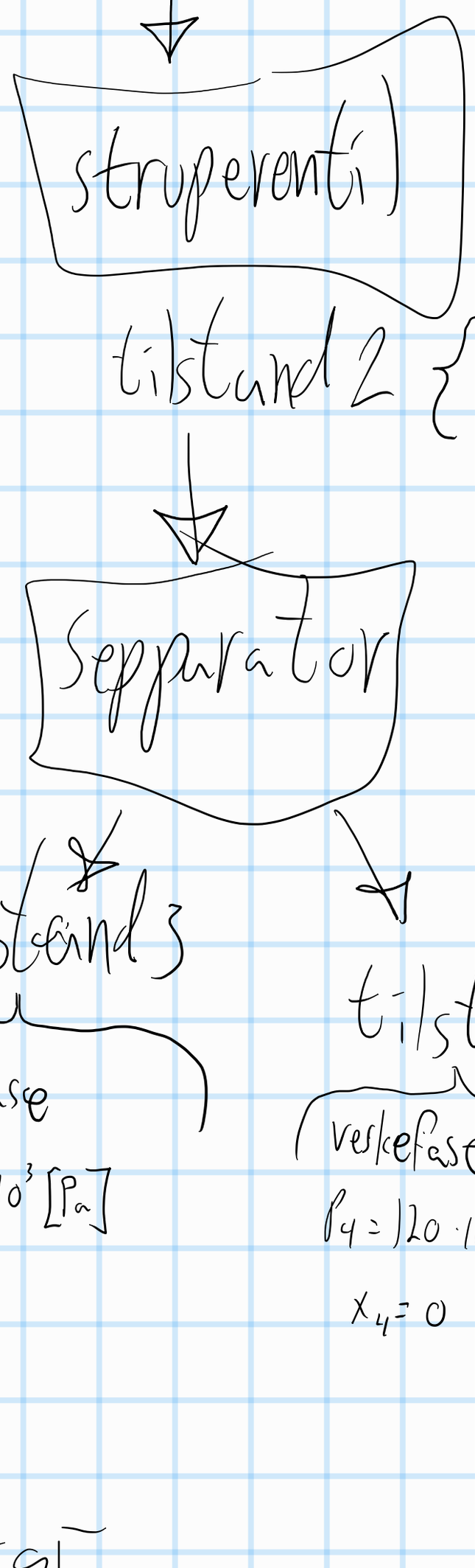
$p_1 > p_{sat@36^\circ C}$

$x_1 = 0 \,, \; V_1 = V_{f@36^\circ C} = 0,0008595$

$\dot{m} = \frac{1}{0,0008595 \cdot 1,5 \cdot 10^{-3}} = \frac{1000}{573} \approx 1,7452 \, [kg/s]$

a)

tilstand | $\{p_1, T_1, \dot{V}_1$



b) strupeventil er isenthalpisk

$$h_2 = h_1$$

$$h_1 = h_{f@36°C} = h_2 = 102,34 \left[\frac{kJ}{kg}\right]$$

trykk er over mettningstrykk

$$+ 0\% \cdot 167,19 \left[\frac{kJ}{kg}\right]$$

$$T_2 = T_{sat@120kPa} = \underline{-22,32 [°C]}$$

$$h_{f@120kPa} = 22,47$$

$$h_{g@120kPa} = 214,52$$

$$x_2 \frac{h_2 - h_{f@120kPa}}{h_{g@120kPa}} = \frac{102,34 - 22,47}{214,52} = \frac{7987}{21452} \approx 0,372$$

$$v_{f@120kPa} = 0,0007523$$

$$v_{g@120kpa} = 0,16216$$

$$V_2 = v_f + x_2(v_g - v_f)$$

$$0,0007323 + 0372\,(0,16216 - 0,0007323)$$

$$v_2 = 0,06 \left[\frac{m^3}{kg}\right]$$

$$\dot{V}_2 = \dot{m} \cdot v_2 = 1,7452 \cdot 0,06 = \underline{\underline{0,104 \left[\frac{m^3}{s}\right]}}$$

c)

Voske → $(\dot{m}_3) = \dot{m} \cdot (1 - x_2) = 1,7452 \cdot (1 - 0,372) = 1,0959856 \left[kg/s \right]$

gass → $(\dot{m}_4) = \dot{m} \cdot x_2 = 1,7452 \cdot 0,372 = 0,6492144 \left[kg/s \right]$

oppg 3

startstate:

$V_1 = 0{,}5\ [m^3]$, $m_1 = 1{,}2\ [kg]$, $T_1 = 27[°C] = 300{,}15[K]$

forsyningsledning:

$T_i = 97[°C] = 370{,}15\ [K]$, $P_i = 500\ kPa$

sluttilstand etter innstrømning

$V_2 = 2 \cdot V_1 = 1\ [m^3]$, $T_2 = 58{,}2[°C] = 331{,}35\ [K]$

omgivelser:

$P_0 = 100\ [kPa]$, $R = 0{,}287\ [kJ/(kg \cdot k)]$

anntar luft behandles som en idealgass.
anntar stempdet beveger seg friksjonsløst og da at trykket en konstant under endring av volum.

$$P_2 = P_1$$

a)

$$P_1 \cdot V_1 = m_1 R \cdot T_1$$

$$P_1 = \frac{m_1 R T_1}{V_1} = \frac{1{,}2 \cdot 0{,}287 \cdot 300{,}15}{0{,}5} = 206{,}74332\ [kPa]$$

b)

$$P_2 = P_1 = 206{,}74332\ [kPa]$$

$$m_2 = \frac{P_2 V_2}{R T_2} = \frac{206{,}74332 \cdot 1}{0{,}287 \cdot 331{,}35} = \frac{206{,}74332}{95{,}097} \approx 2{,}174\ [kg]$$

$$\Delta m = m_2 - m_1 = 2{,}174 - 1{,}2 = 0{,}974\ [kg]$$

c)

$$W_0 = P_0 (V_2 - V_1) = 100(1 - 0,5) = 50\ [kJ]$$

$$W_t = P_1 (V_2 - V_1) = 206,74332 \cdot 0,5 = \underline{103,37166}$$

$$W_{nyttig} = W_t - W_0 = 103,37166 - 50 = \underline{\underline{53,37166}}$$

d)

$$V_3 = 0,5\ [m^3]$$

$$\frac{T_3}{T_2} = \left(\frac{V_3}{V_2}\right)^{k-1} \qquad k = \frac{C_p}{C_v}$$

$$C_p = 1,005 \left[\frac{kJ}{(kg \cdot k)}\right]$$

$$C_v = 0,718 \left[\frac{kJ}{(kg \cdot k)}\right]$$

$k = 1,4$ ← lest fra tabell

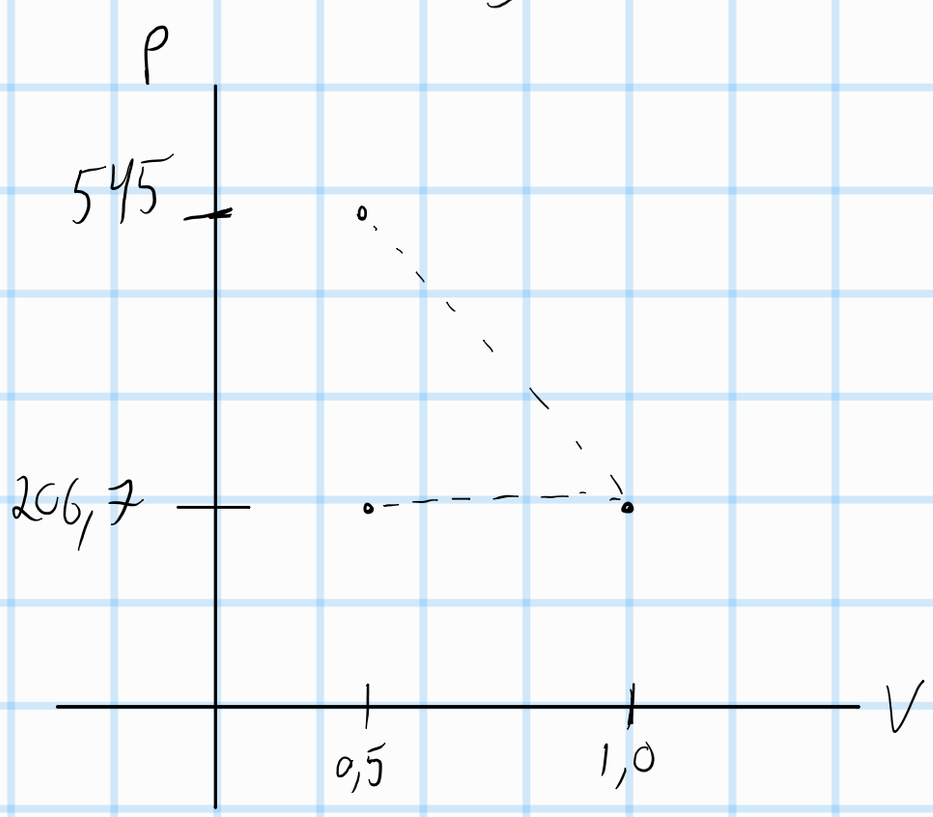
$$T_3 = \left(\frac{1,0}{0,5}\right)^{\frac{0,4}{(1,4-1)}} \cdot 331,35 \approx 437,21 \; [K]$$

e)

$$p_3 V_3 = m R T_3 \rightarrow mR = \frac{p_3 V_3}{T_3}$$

$$p_2 V_2 = m R T_2 \rightarrow mR = \frac{p_2 V_2}{T_2}$$

$$p_3 = \frac{T_3}{V_3} \frac{p_2 V_2}{T_2} = \frac{437,21}{0,5} \frac{206,74332 \cdot 1,0}{331,35} = 545,5877286 \; [kPa]$$



oppg 4

temp i heat source $\left(T_H\right) = 1150[^oC] = 1423,15[K]$

termic virkningsgrad $\left(\eta_{th}\right) = 42\% = 0,42$

maksimal utført arbeid per syklus $\left(W_n\right) = 550\,[kJ]$

a)

nødvendig varmemengde $\left(\dot{Q}_H\right) = \dfrac{W_n}{\eta_{th}} = \dfrac{550}{0,42} \approx \underline{\underline{1309,52[kJ]}}$

b)

spillvarmemengden per syklus $\left(\dot{Q}_L\right) = \dot{Q}_H - W_n = 1309,52 - 550$

$= 759,52\,[kJ]$

c)

varmesluket (T_L)

$$\frac{T_L}{T_H} = 1 - \eta_{th} \rightarrow T_L = T_H(1 - \eta_{th}) = 1423,15(1 - 0,42)$$

$$= 825,427 [K]$$